1. In this argument, the writer made the mistake of assuming that *any* set of n horses are the same color. In reality, it should be that the first n horses are the same color. With that condition, the writer cannot proceed to say that H_2, ..., H_{n+1} are the same color. □

2. The sum of the first n odd numbers is n^2. To prove this, we can start with the base case, which is n = 1. Since it is the first odd number, the sum on the left is equal to 1, and 1^2 is also 1. Therefore, the statement is true when n = 1.

   Now, we can do the inductive step. Let's assume that 1 + 3 + 5 + … + 2n - 1 = n^2. We need to prove that 1 + 3 + … + 2n - 1 + 2n + 1 = $(n + 1)^2$. We can replace 1 + 3 + … + 2n - 1 with n^2 and expand $(n + 1)^2$ on the right to get $n^2 + 2n + 1 = n^2 + 2n + 1$, which we know is true. Therefore, if it is true for n, it is true for n + 1, so it is true for all positive integers n. □

3. We can rewrite the statement using Σ notation as $\sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4} = (\sum_{i=1}^{n} i)^2$

   Let's start with the right-hand side. We know that the expression $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$, so squaring the equation gives us the desired result.

   Now, for the left-hand side, we start with the base case of n = 1. We see that the equation is true, as $1^2(2)^2/4 = 1$.

   For the inductive step, let's assume that $\sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}$. We need to prove that $\sum_{i=1}^{n+1} i^3 = \frac{(n+1)^2(n+2)^2}{4}$. We can rewrite that $\sum_{i=1}^{n} i^3 + (n+1)^3 = \frac{(n+1)^2(n+2)^2}{4}$, and then replace $\sum_{i=1}^{n} i^3$ with $\frac{n^2(n+1)^2}{4}$ to get $\frac{n^2(n+1)^2}{4} + (n+1)^3 = \frac{(n+1)^2(n+2)^2}{4}$. We can subtract $\frac{n^2(n+1)^2}{4}$ from both sides and then simplify the right side further, and we end up with $(n+1)^3 = (n+1)^3$, which we know is true. □

4. We need to prove that $n^5 - n = 5k$ for all nonnegative integers n, where k is an integer. The base case of n = 1 gives us k = 0, so it works for n = 1.

   Now, we can do the inductive step by assuming $n^5 - n = 5k_0$. That means we have to prove that $(n+1)^5 - (n+1) = 5k$. Expanding and simplifying the left side of the equation, we get $n^5 + 5n^4 + 10n^3 + 10n^2 + 4n = 5k$, or $(n^5 - n) + 5n^4 + 10n^3 + 10n^2 + 5n = 5k$. We can replace $(n^5 - n)$ with $5k_0$ and then divide both sides by 5 to get $n^4 + 2n^3 + 2n^2 + n = k - k_0$. We know that is true,

because the left side of the equation is an integer. □

5. We know this is true for n = 0, because $1 \geq 1$. Now, let's assume $(1 + x)^n \geq 1 + nx$. We need to prove that $(1 + x)^{n+1} \geq 1 + (n + 1)x = 1 + x + nx$. We know that $(1 + x)^{n+1} = (1 + x)^n(1 + x) \geq (1 + nx)(1 + x) = 1 + x + nx + nx^2$. Since $(1 + x)^{n+1} \geq 1 + x + nx + nx^2$, it must also be greater than or equal to $1 + x + nx$, because $nx^2 \geq 0$. □

6. First, let's show this works for n = 0 and n = 1. If we plug in n = 0, we get $\frac{1}{\sqrt{5}}(1 - 1) = 0$, and if we plug in n = 1, we get $\frac{1}{\sqrt{5}}(\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2}) = \frac{1}{\sqrt{5}}(\sqrt{5}) = 1$.

   Now, we need to prove that if $F_n = \frac{1}{\sqrt{5}}((\frac{1+\sqrt{5}}{2})^n - (\frac{1-\sqrt{5}}{2})^n)$ and $F_{n+1} = \frac{1}{\sqrt{5}}((\frac{1+\sqrt{5}}{2})^{n+1} - (\frac{1-\sqrt{5}}{2})^{n+1})$, then $F_{n+2} = \frac{1}{\sqrt{5}}((\frac{1+\sqrt{5}}{2})^{n+2} - (\frac{1-\sqrt{5}}{2})^{n+2})$. Since $F_{n+2} = F_{n+1} + F_n$, we must prove that $(\frac{1+\sqrt{5}}{2})^{n+2} - (\frac{1-\sqrt{5}}{2})^{n+2} = (\frac{1+\sqrt{5}}{2})^{n+1} - (\frac{1-\sqrt{5}}{2})^{n+1} + (\frac{1+\sqrt{5}}{2})^n - (\frac{1-\sqrt{5}}{2})^n$. We can rewrite that as $(1 + \sqrt{5})^n((\frac{1+\sqrt{5}}{2})^2 - (\frac{1+\sqrt{5}}{2}) - 1) = (1 - \sqrt{5})^n((\frac{1-\sqrt{5}}{2})^2 - (\frac{1-\sqrt{5}}{2}) - 1)$. We can also find that $(\frac{1+\sqrt{5}}{2})^2 - (\frac{1+\sqrt{5}}{2}) - 1 = 0$ and $(\frac{1-\sqrt{5}}{2})^2 - (\frac{1-\sqrt{5}}{2}) - 1 = 0$. That means both sides are equal to 0. □

7. First, we start with the base case of n = 1. We can say k = 1 and $a_1 = 1$, and we get $\sum_{i=1}^{k} a_i \cdot i! = 1$.

   For the inductive step, we start by assuming $n = \sum_{i=1}^{k_0} a_i \cdot i!$. We need to prove that $n + 1 = \sum_{i=1}^{k} b_i \cdot i!$ where $0 \leq b_i \leq i$. If we replace n with $\sum_{i=1}^{k_0} a_i \cdot i!$, we get $1 = \sum_{i=1}^{k} b_i \cdot i! - \sum_{i=1}^{k_0} a_i \cdot i!$. We also know that $1 = \sum_{i=1}^{1} 1 \cdot i!$, so to make the equation true, we can say $k = k_0 + 1$. Also, b should have k elements, a should have k - 1 elements, and the elements of a should be the first k - 1 elements in b. If those conditions are satisfied, then $n + 1 = \sum_{i=1}^{k} b_i \cdot i!$. □

8. We can restate the problem as $a\sum_{i=1}^{3^n} 10^{i-1} = k \cdot 3^n$ where a is a positive integer less than 10 and n and k are positive integers. To prove that is true, we start with the base case of n = 1. That gives us a(111) = 3k, or k = 37a.

   For the inductive step, we assume that $a\sum_{i=1}^{3^n} 10^{i-1} = k_0 \cdot 3^n$, and then we prove that $a\sum_{i=1}^{3^{n+1}} 10^{i-1} = k \cdot 3^{n+1}$. We can rewrite the left-hand side of the equation as $a\sum_{i=1}^{3^n} 10^{i-1} + a \cdot 3^{n+1} \cdot 10^{3^n}$, or $k_0 \cdot 3^n + a \cdot 3^{n+1} \cdot 10^{3^n}$. So we get $3^n(k_0 + 3a \cdot 10^{3^n}) = 3^n(3k)$, or $k_0 + 3a \cdot 10^{3^n} = 3k$. So if k_0 is a multiple of 3, then the equation would be true. □

9. We can rewrite the problem as $\sum_{i=1}^{n} a_i \cdot 10^{i-1} = k \cdot 5^n$ where n and k are positive integers and a is a list of n single digit odd numbers. The base case, n = 1, is true, because the number 5 is a one digit odd number that is divisible by 5^1.

   Now, we do the inductive step by assuming that $\sum_{i=1}^{n} a_i \cdot 10^{i-1} = k_0 \cdot 5^n$ is true, and then proving that $\sum_{i=1}^{n+1} b_i \cdot 10^{i-1} = k \cdot 5^{n+1}$. Rewriting the left side of the equation, we get $\sum_{i=1}^{n+1} b_i \cdot 10^{i-1} = \sum_{i=1}^{n} b_i \cdot 10^{i-1} + b_{n+1} \cdot 10^n$. If we assume the first n digits of b are the same as those of a, we get $k_0 \cdot 5^n + b_{n+1} \cdot 10^n = k \cdot 5^{n+1}$, or $k_0 + b_{n+1} \cdot 2^n = 5k$. Now, if k_0 and b_{n+1} are both multiples of 5, the equation is true. □

10. First, we start with the base cases of n = 1 and n = 2. Those are just the 2nd and 3rd Fibonacci numbers, so they are the sum of nonconsecutive Fibonacci numbers.

    For the inductive step, we assume that it holds for all positive integers up to some positive integer k, and then prove that it holds for k + 1. We can split this into two cases:

    Case 1: $F_i = k + 1$
    In this case, k + 1 is a Fibonacci number, so our claim holds.

    Case 2: $F_i < k + 1$
    Let's say F_i is the largest Fibonacci number less than k + 1. That means $k + 1 < 2F_i$,

because F_{i+1} is the sum of F_i and a number smaller than F_i, and since $k + 1 < F_{i+1}$, it must be smaller than $2F_i$ as well. We can rewrite that as $k + 1 - F_i < F_i$. Since that is an integer less than F_i, it can be written as the sum of nonconsecutive Fibonacci numbers, and since F_i is the largest Fibonacci number less than $k + 1$, our claim holds. □

11. Here are the formulas and their proofs for each of m = 0, 1, 2, 3, and 4:
    a. A(0, n) = n + 1
       To prove this, we can look back at the definition of the Ackermann function, which states that when m = 0, A(m, n) = n + 1. That means A(0, n) = n + 1. □

    b. A(1, n) = n + 2
       To prove this, we can start with the base case, n = 0. By the definition of the Ackermann function, that gives us A(1, 0) = A(0, 1) = 1 + 1 = 2.

       For the inductive step, let's assume A(1, k) = k + 2 for some nonnegative integer k, and show that A(1, k + 1) = k + 3. We know that A(1, k + 1) = A(0, k + 2) by the inductive assumption, and that A(0, k + 2) = k + 2 + 1 = k + 3. Therefore, A(1, k + 1) = k + 3. □

    c. A(2, n) = 2n + 3
       To prove this, we start with the base case: n = 0. That gives us A(2, 0) = A(1, 1) = A(0, A(1, 0)) = A(0, 2) = 3, as desired.

       Now, we can do the inductive step. First, we assume that A(2, k) = 2k + 3, and then prove that A(2, k + 1) = 2k + 5. By the definition of the Ackermann function and the statement proved in the previous part, A(2, k + 1) = A(1, A(2, k)) = A(1, 2k + 3) = 2k + 5. Therefore, A(2, k + 1) = 2k + 5. □

    d. A(3, n) = $2^{n+3} - 3$
       Let's start with the base case, n = 0. A(3, 0) = A(2, 1) = 2(1) + 3 = 5. Also, 2^3 - 3 = 5.

       Now, we assume that A(3, k) = $2^{(k+3)}$ - 3 for some nonnegative integer k. We want to prove that A(3, k + 1) = 2^{k+4} - 3. Using the definition of the Ackermann function, A(3, k + 1) = A(2, A(3, k)) = A(2, 2^{k+3} - 3) = 2(2^{k+3} - 3) + 3 = 2^{k+4} - 3. So A(3, k + 1) = 2^{k+4} - 3. □

    e. I couldn't find an explicit formula for A(4, n), but I managed to find a recursive formula:

       A(4, n) = $2^{A(4, n-1)+3}$ - 3; A(4, 0) = 13, A(4, 1) = 2^{16} - 3, A(4, 2) = $2^{2^{16}-3} - 3$, etc.

12. Proofs:

a. Every coloring of G\e either works for G as well or it doesn't. If it doesn't work, then the two vertices connected by e in graph G are the same color. But those are the same colorings that would work when e is contracted. So all the proper colorings of G are the same ones in G\e, except for the ones in G/e. Rewriting that, we get $C_G(n) = C_{G\backslash e}(n) - C_{G/e}(n)$. □

b. First, we start with the base case, where there are 0 vertices and 0 edges. That means $C_G(n) = 0$, which is a polynomial.

   Now, for the inductive step, we can assume that $C_G(n)$ is a polynomial function for all graphs where the sum of the number of edges and vertices is less than or equal to k. Now, let's consider a graph G with k + 1 edges plus vertices.

   First, let's consider the case where we remove one vertex and no edges by contracting that edge. The number of ways to color it is $C_{G/e}(n)$. Since G/e has less vertices plus edges than G, $C_{G/e}(n)$ is a polynomial function. Now if we consider graph G, we see that $C_G(n) = (n - a)*C_{G/e}(n)$, where a is the number of colors the vertex can't be. Since multiplying a polynomial by (n - a) also results in a polynomial, $C_G(n)$ is also a polynomial function.

   Now, let's consider the case where we remove one edge and no vertices. By doing that, we get a graph (which we can call H) where $C_H(n)$ is a polynomial function of n. By adding the edge back, we are essentially either changing the factor of (n - a) in $C_H(n)$ to (n - a - 1) or keeping it the same, because one of the vertices at the end of the edge now has one less option to choose from if it is different from the other vertices that connects to. If it is the same color, then it stays the same as (n - a). So $C_G(n)$ is just the same as $C_H(n)$, but (n - a) is potentially changed to (n - a - 1). That means $C_G(n)$ is a polynomial function. □

c. The function $C_G(n)$ is just the product of the number of ways to color each vertex such that no two vertices connected by an edge are the same color. A vertex has n - a options, where a is the number of colors that the vertex can't be. That means, if there are k vertices, $C_G(n) = (n - a_1)(n - a_2)...(n - a_k)$, where a_i is a constant for $1 \leq i \leq k$. Since there are k terms being multiplied, the degree of $C_G(n)$ is k. □

d. By Vieta's formulas, $\sum_{1 \leq i_1 \leq i_2 \leq i_3 \cdots i_k \leq n} \left(\prod_{j=1}^{k} r_{i_j}\right) = (-1)^k \frac{a_{n-k}}{a_n}$. Basically, this means that the sum of multiple products of the roots of the polynomial is $(-1)^k \frac{a_{n-k}}{a_n}$. In our case, all the roots of $C_G(n) = (n - a_1)(n - a_2)...(n - a_k)$ are positive, so $(-1)^k \frac{a_{n-k}}{a_n}$ is positive. We also know that a_n is 1. So $(-1)^k a_{n-k}$ is positive. When k = 0, we get

a_n is positive, when k = 1, we get a_{n-1} is negative, when k = 2, we get a_{n-2} is positive, etc. That means the signs of the coefficients of $C_G(n)$ alternate. □